

E-Waste Segregation using GAN

R PUNITHA
Assistant Professor
Department of Information
Technology
Sri Sai Ram Institute of
Technology
Chennai, India
punitha.it@sairamit.edu.in

K ANURATHA
Associate Professor
Department of Information
Technology
Sri Sai Ram Institute of
Technology
Chennai, India
anuratha.it@sairamit.edu.in

S ANITHA JEBAMANI
Associate Professor
Department of Information
Technology
Sri Sai Ram Engineering
College
Chennai, India
jebamani.it@sairam.edu.in

KEERTHANA PRIYA U
*Department of Information
Technology*
Sri Sai Ram Institute of
Technology
Chennai, India
sit22it071@sairamtap.edu.in

LAKSHYA S
*Department of Information
Technology*
Sri Sai Ram Institute of
Technology
Chennai, India
sit22it169@sairamtap.edu.in

MRINAALINI S
*Department of Information
Technology*
Sri Sai Ram Institute of
Technology
Chennai, India
sit3t24it2@sairamtap.edu.in

Abstract: E-waste management has emerged as a pressing global environmental challenge, driven by the rapid growth of technology and the short lifecycle of electronic devices. The improper disposal of electronic waste not only leads to resource loss but also releases toxic substances, posing serious risks to the environment and human health. To address this issue, this paper proposes an innovative automated e-waste segregation system that integrates both hardware and software components for efficient sorting, detection, and classification. The process begins with a proximity sensor, interfaced with an Arduino microcontroller, which detects metallic objects and separates them from non-metal waste. This ensures that recyclable metals are handled first. Once the metallic waste is isolated, it is passed to the next stage, where a Generative Adversarial Network (GAN) model is employed for advanced classification. The GAN, trained on an extensive dataset of e-waste images, analyses the metal waste to identify specific electronic components such as circuit boards, wires, and connectors. This multi-stage approach reduces manual effort, increases classification accuracy, and promotes sustainable e-waste recycling practices.

Index Terms – *E-waste Management, Sensor, Generative Adversarial Network (GAN), Automated Waste Segregation, Machine Learning, Waste Classification, Image recognition, Hazardous Substances, Real-Time Detection.*

I. INTRODUCTION

Machine Learning (ML) can play a role in automating the segregation of e-waste. Image recognition algorithms, a subset of ML, can be employed to identify and categorize electronic wastes based on images. Electronic waste is one of the fastest-growing waste streams, with India generating about 3.2 million metric tons annually, ranking third globally. Improper disposal releases toxic substances like lead, mercury, and cadmium, posing serious environmental and health risks. This highlights the urgent need for automated and efficient systems to segregate and process e-waste. By using ML and advanced AI algorithms, this paper proposes a scalable and sustainable solution for modern waste management facilities, addressing the pressing need for automated e-waste segregation to promote environmental sustainability and responsible recycling practices. The Arduino microcontroller coordinates the initial sorting process, controlling the sensors and mechanical actuators for real-time waste segregation. Simultaneously, the GAN module processes visual data from the metal waste and distinguishes between e-waste and metals. The seamless communication between the microcontroller and the GAN model ensures efficient decision-making and automation throughout the process.

The key objectives of this research are:

A. Develop an intelligent system that accurately classifies and separates e-waste from other materials using Arduino sensors and GAN.

B. Improve the recovery of valuable electronic components and metals, reducing environmental pollution.

C. Reduce human intervention in waste segregation, making the process faster, safer, and more reliable.

D. Prevent hazardous e-waste materials from contaminating landfills and water bodies by enabling proper recycling.

E. Utilize a trained GAN model to differentiate between electronic waste and metal waste with high precision.

II. RELATED WORKS

In [1], This study proposes an IoT-based smart waste management system using deep learning for automated segregation and monitoring. A quantized SSD MobileNetV2 model on Raspberry Pi 4 classifies waste into five types, while a camera and servo motor handle detection and sorting. Sensors track bin levels, GPS provides location, LoRa transmits status, and RFID ensures secure access. In [2], This paper proposes a thermal imaging method to classify e-waste into metal, plastic, glass, and PCB. Using long-wave infrared (8–15 μm) thermograms for feature extraction, the model achieved 84–96% accuracy on simulated e-waste, indicating strong potential for scalable automated recycling. In [3], This study applies Deep Neural Networks (DNN) in TensorFlow, using Python, to classify images from a five-category flower dataset. Achieving over 90% average accuracy, with roses at 90.585%, the results highlight DNN's effectiveness for flower image recognition. In [4], This system employs Capsule Neural Networks to classify plastic and non-plastic waste, achieving 96.3% and 95.7% accuracy on public and private datasets, outperforming traditional CNNs. A ResNet50-SVM model reached 87% accuracy. A hardware prototype validates its practicality for low-cost, safe waste management. In [5], This paper proposes an IoT-based smart waste segregation system for real-time detection and classification. A microcontroller integrates sensors for waste presence, moisture, and metal detection, while image processing distinguishes plastic from biodegradable waste. Bin status is uploaded to the cloud for efficient monitoring and disposal. In [6], This paper presents an automated e-waste detection, sorting, and segregation system to enhance recycling. A conveyor mechanism

separates electronic from non-electronic waste, followed by shredding to recover reusable materials. A web-based platform enables digital collection and real-time monitoring for efficient e-waste management. In [7], This paper presents ElectroSortNet, a CNN-based model for e-waste classification, enhanced with squeeze and excitation attention and residual connections. Trained on a novel dataset, it achieved 97.44% accuracy, outperforming prior architectures. An IoT-driven system integrating ElectroSortNet with a conveyor belt and robotic arms enables automated e-waste segregation. In [8], The proposed system integrates a camera and a weighing scale. Using object detection, e-waste items are classified based on components such as circuit boards, sensors, and wires. Simultaneously, the scale estimates each component's average value relative to its weight. In [9], This paper proposes a system that identifies e-waste products, lists their components, suggests suitable recycling methods, and recommends appropriate recycling facilities. Leveraging deep learning, the approach enhances the efficiency of segregation, classification, and recycling processes. By analyzing the composition of diverse e-waste items, the system provides optimized recycling recommendations, supporting sustainable e-waste management. In [10], This study develops a YOLOv7-based e-waste segregation system aligned with the WEEE directive, featuring motor control and a GUI for bin monitoring. It achieves 93.75% precision for ferrous metals, 94.00% recall for others, an F1 score of 91.84%, and 88.40% overall accuracy. In [11], This study presents an autonomous waste segregation system using sensors, including an A88 metal proximity sensor, a customized moisture sensor, an IR sensor, and an ultrasonic sensor. The system effectively classifies waste based on its properties, promoting reuse and recycling for creating new goods. In [12], This system uses computer vision and machine learning to automate waste segregation, sorting materials into biodegradable, recyclable, reusable, and trash categories in real time. A camera captures images of incoming waste, which are processed by a YOLOv8 deep learning model for accurate classification. The waste is then directed to its respective collection point. This approach offers a cost-effective, eco-friendly solution that supports sustainable waste management.

III. IMPLEMENTATION

This research presents the design and implementation of an automated e-waste segregation system that identifies and separates electronic waste from general waste using a combination of sensors, microcontrollers, and a GAN based deep learning model.

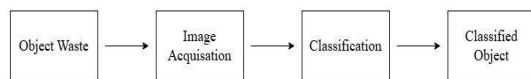


Fig 1. Process of waste segregation

Figure 1, shows the waste segregation process that employs GAN, a machine learning model to classify and separate e-waste. Using the proximity sensor, the metal wastes are separated from other wastes. The discriminator is fed with the datasets of e – wastes beforehand. The images of the metal wastes that are separated from other wastes are acquired by the generator. The discriminator distinguishes between its fed dataset and generator captured image. If the captured image matches with the dataset, then it is classified as e – waste. Then the servo motor is run to put the wastes in their respective bins.

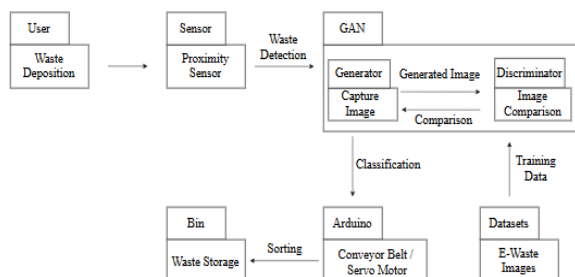


Fig 2. Architecture Diagram

Figure 2, depicts the architecture diagram, which provides the system overview of the e- waste segregation system.

A. Waste Input and Detection

The system begins with mixed waste materials being placed onto an input tray or a conveyor belt. A proximity sensor is positioned to detect metallic content in the objects. When a metal- containing item is detected, the sensor sends a signal to the Arduino microcontroller. This allows the system to automatically separate metallic waste from non-metallic waste at an early stage. Non-metal items are diverted away, while metal items continue to the next stage for further classification.

B. Initial Sorting by Arduino

The Arduino acts as the main control unit, processing inputs from the proximity sensor. When a metallic object is detected, the Arduino activates an actuator, such as a servo motor or solenoid, to move the item onto a secondary conveyor or designated path. If the object does not contain metal, it is automatically excluded from further processing. This step ensures that only potentially recyclable metal items are analyzed for e-waste content.

C. Image Capture for Metal Waste

Once the item is confirmed as metallic, it moves to a section of the system equipped with a camera or vision module. This module captures a clear image of the item as it passes by. The image is essential for the next phase of classification, where artificial intelligence is used to determine if the item is an electronic component or a simple piece of metal.

D. E-Waste Classification Using GAN

The captured image is processed using a Generative Adversarial Network (GAN), which has been trained to distinguish between e-waste (like circuit boards, capacitors, or wires) and other metallic objects (such as screws, nails, or cans). The GAN consists of a generator and a discriminator, where the discriminator is used here to classify the object based on visual features. The model outputs a classification result indicating whether the item is e-waste or not which is then sent back to the Arduino for the next action.

Preprocessing Module: This module prepares the captured images of waste objects for analysis. It involves resizing, noise removal, contrast enhancement, and normalization to ensure consistency and improve the model's accuracy during classification.

Segmentation Module: The segmented module isolates the actual object from the background in the image. This step enhances focus on the object's shape, texture, and structure, helping the model distinguish between e-waste and general metal components.

Classification Module: In this final stage, the trained discriminator of the GAN model analyses the segmented image and classifies it as e-waste or

non-electronic metal waste based on learned features. The result is then sent to the Arduino to control sorting.

E. Final Sorting and Actuation

Based on the classification result from the GAN, the Arduino triggers an appropriate mechanical actuator to direct the object to the correct bin. For instance, if the item is identified as e-waste, it is moved to a dedicated bin for electronic components. If it is general metal waste, it is routed to a separate bin. This is achieved using various actuator mechanisms such as flaps, arms, or conveyors that respond in real-time to the output.

The key tools of this system are as follows:

1. Proximity Sensor: Detects metal waste and triggers an actuator to separate metallic waste from other materials.
2. Arduino: Acts as the central microcontroller for sensor inputs and actuates motors for waste separation.
3. GAN (Generative Adversarial Network): Classifies the wastes into e- waste and other waste types.
4. Actuators and Conveyor Belts: Mechanically separate identified waste based on classifications.
5. MATLAB IDE: The main development environment used to write, test, and debug both the Arduino control code and the deep learning model.
6. MATLAB Deep Learning Toolbox: For developing and training the GAN module.
7. MATLAB Image Processing Toolbox: For preprocessing images before training the GAN model.

IV. RESULTS AND ANALYSIS

In tackling the challenge of e-waste segregation, our goal was to go beyond traditional methods and build a system that is intelligent, precise, and adaptable. To achieve this, we chose to implement a GAN a powerful deep learning model known for its exceptional ability to understand and generate complex visual patterns. Unlike standard image classifiers, GANs are trained through a competitive learning process that makes them particularly effective at identifying fine-grained visual differences, which is critical when distinguishing between visually similar items like electronic components and ordinary metal waste.

Inference time refers to how long the trained model takes to identify the waste type from the camera feed. A lower inference time indicates faster detection. The proposed model has achieved a smaller inference time when compared to the existing system using CNN based waste classification.

MATLAB has been employed as the primary tool for preprocessing, segmenting, and classifying e-waste images. Leveraging its computational capabilities, an efficient workflow has been developed to process large datasets, extract essential features, and apply GAN to accurately categorize different types of wastes.

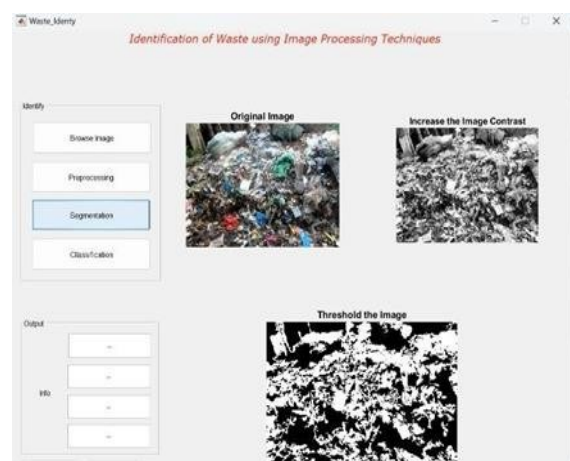


Fig 3. The segmented image for feature extraction.

Figure 3, represents the preprocessing and segmentation steps in waste identification. The original waste image is enhanced by increasing contrast, followed by thresholding to create a binary segmented image that highlights the boundaries of different waste materials for further classification shows Figure 4.

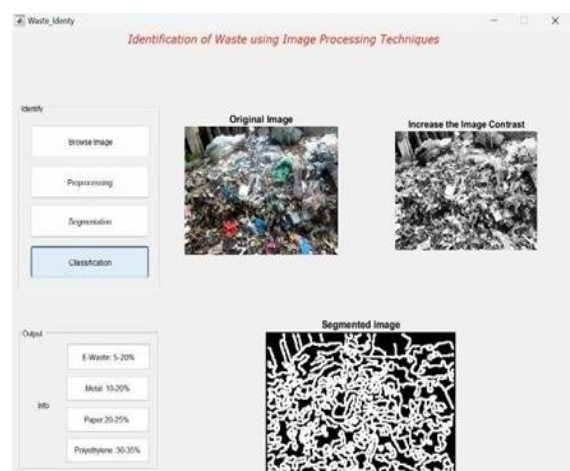


Fig 4. The classification result displaying the percentage composition of e-waste.

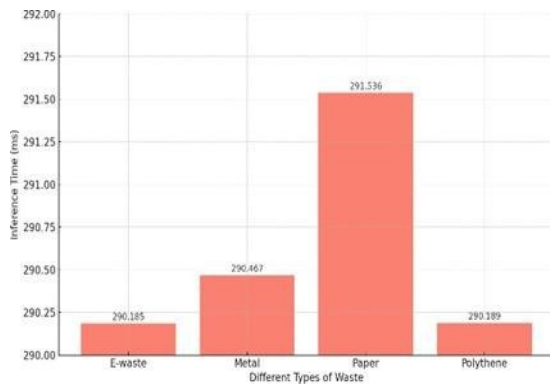


Fig 5. Inference time analysis (GAN).

Figure 5, shows the bar graph representing the inference time of different wastes classified through GAN.

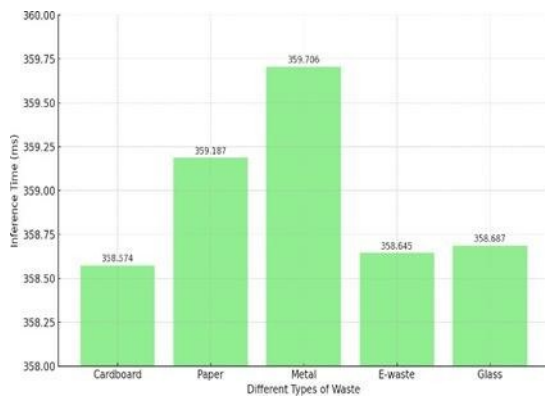


Fig 6. Inference time analysis (CNN).

Figure 6, shows the inference time of different types of wastes classified by “A CNN-Based Smart Waste Management System Using TensorFlow Lite and LoRa-GPS Shield in Internet of Things Environment”.

In the landscape of waste management technologies, many systems still rely on basic sensors or manual sorting. Our project goes beyond, offering a level of automation and precision that surpasses existing methods.

Advanced Classification Using AI: While many existing waste segregation systems use simple sensors or rule-based logic, our project integrates deep learning with GANs, enabling intelligent decision-making based on image features. **Dual-Stage Sorting:** The system doesn't just detect metal, it goes further to differentiate between e-waste and regular metal, making it more precise and suitable for recycling industries. **Automation and Safety:** Reduces the need for human intervention in hazardous e-waste handling, increasing safety and efficiency compared to manual or semi-automated systems. **Cost-Effective and Scalable:** Built using Arduino, the system is low-cost and scalable. The AI module can be retrained for other types of waste, making it adaptable to various environmental needs.

V. CONCLUSION

The E-waste segregation system integrating Arduino control, inductive sensing, and GAN-based image classification demonstrates a promising approach for automating the separation of electronic waste from general metallic waste streams. By combining early-stage metal detection with high-accuracy vision-based classification, the system improves sorting efficiency, reduces human intervention, and supports environmentally responsible waste handling by preventing harmful materials from entering landfills. The approach is eco-friendly, as it produces no hazardous emissions or byproducts, and enables the recovery of valuable components for recycling. However, practical limitations remain such as the system's sensitivity to lighting variations, motion blur at higher conveyor speeds, and inability of inductive sensors to detect non-metallic e-waste items. Additionally, the GAN model's performance is dependent on the diversity and quality of its training dataset, which may be constrained by limited labelled samples. Future improvements could include integrating multi-sensor fusion to detect plastic-rich or mixed-material e-waste, developing a multi-class classification scheme for more granular sorting, and implementing real-time adaptive lighting control to mitigate environmental variations. Furthermore, leveraging GANs for synthetic data generation could enhance dataset size and diversity, improve the robustness of classification models when the labelled data is scarce. The scalability of the design could be explored for industrial-scale deployments, incorporating faster processing hardware and modular conveyor units. By addressing these challenges, the system could evolve into a comprehensive, intelligent waste management solution adaptable to various waste types, significantly contributing to sustainable recycling practices and a toxin-free environment.

VI. REFERENCES

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